Short-Term Forecasting of South Africa's Hourly Electricity Demand Using Machine Learning

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Abstract

Accurate short-term demand forecasts are essential for scheduling generation and limiting supply shortfalls on South Africa's constrained national grid. This study evaluates statistical and machine learning methods for forecasting hourly RSA contracted demand 24 hours ahead, using 40,248 hourly observations from the Eskom Data Portal covering 1 April 2022 to 2 November 2026. Features describing the daily, weekly and annual cycles were constructed with Fourier terms, together with calendar, public holiday, load shedding and lagged-demand variables. Two naive benchmarks, a Fourier regression estimated by ordinary least squares, a random forest and an extreme gradient boosting (XGBoost) model were trained on data up to 30 April 2026 and evaluated on a hold-out period from 1 May to 2 November 2026. To assess the effect of load shedding on model estimation, each model was trained both on the full sample and on data from April 2024 onwards only. XGBoost trained on the post-April 2024 sample achieved the lowest error, with a mean absolute percentage error (MAPE) of 2.30%, compared with 2.40% for Eskom's published forecast and 3.50% for a naive same-hour-previous-day forecast. During the evening peak (17:00 to 20:59) its mean absolute error was 462 MW, 14% lower than that of Eskom's forecast. Demand at the same hour one week earlier was the most influential predictor. Exploratory analysis also revealed a decline in typical daily demand from approximately 26 to 27 GW in 2022 to 22 GW in 2026. Because the models were evaluated as rolling 24-hour-ahead forecasts over a single test period, the results indicate performance comparable to, rather than conclusively better than, Eskom's operational forecast.

Keywords: short-term load forecasting; electricity demand; machine learning; XGBoost; Fourier regression; load shedding; South Africa

1 Introduction

1.1 Background

Electricity cannot be stored economically at grid scale, so a power system must balance supply and demand continuously. Short-term load forecasts, with horizons from one hour to several days, underpin unit commitment, generation dispatch and reserve planning [1, 2]. Underestimating demand risks supply shortfalls, while overestimating it leads to the unnecessary use of costly peaking plant.

In South Africa these consequences are acute. Between 2022 and early 2024 the national grid experienced its most severe load shedding on record, as constrained generation capacity was repeatedly unable to meet demand. At the same time, the rapid growth of privately owned embedded generation, mainly rooftop solar photovoltaic (PV) systems, has reduced the demand served by the national utility. The Council for Scientific and Industrial Research (CSIR) reported that energy demand fell by about 3% between 2023 and 2024 and attributed the decline mainly to this embedded generation, while Eskom estimated that more than 6 GW of private solar PV had been installed by September 2024 [3]. Demand forecasting in South Africa therefore faces both strong regular cycles and structural change.

1.2 Problem statement

Eskom publishes an hourly demand forecast, but independent, reproducible evaluations of modern machine learning methods on recent South African hourly data remain scarce. Much of the published South African work forecasts daily peak demand using data from before the recent load shedding crisis and the growth of embedded solar generation. It is therefore unclear how well current machine learning methods forecast hourly national demand under these new conditions, how they compare with the operator's own forecast, and whether data from the load shedding period help or harm model estimation.

1.3 Aim

The aim of this study is to develop and evaluate models that forecast South Africa's hourly electricity demand 24 hours ahead, and to compare their accuracy with Eskom's published forecast.

1.4 The specific objectives are to:

1. obtain and pre-process more than four years of hourly demand data from the Eskom Data Portal;
2. characterise the daily, weekly and seasonal structure of demand and the effect of load shedding;
3. construct predictors describing demand cycles, calendar effects and recent demand;
4. estimate a Fourier regression, a random forest and an XGBoost model, and compare them with naive benchmarks and Eskom's forecast; and
5. assess whether excluding the period of heavy load shedding from training improves accuracy.

1.5 Research questions

1. How accurately can machine learning models forecast South Africa's hourly electricity demand 24 hours ahead?
2. How does their accuracy compare with naive benchmarks and with Eskom's published forecast, overall and during the evening peak?
3. Does training only on data after the period of heavy load shedding improve forecast accuracy?
4. Which predictors contribute most to forecast accuracy?

1.6 Significance of the study

The study provides a reproducible benchmark for hourly demand forecasting on recent South African data, built entirely on publicly available data and open-source software. Its findings on the evening peak, the period of greatest stress on the grid, and on the treatment of load shedding data are relevant to system planners and to future forecasting research in South Africa.

1.7 Organisation of the paper

Section 2 reviews related literature. Section 3 describes the data and their pre-processing, and Section 4 sets out the methodology. Section 5 presents the results, Section 6 discusses them with reference to the research questions, and Section 7 concludes.

2 Literature review

2.1 Approaches to short-term load forecasting

Short-term load forecasting has a long history, and the methods used fall broadly into statistical time-series models and machine learning models [1, 2]. Electricity demand typically exhibits several seasonal cycles at once. Taylor [4] showed that half-hourly demand contains both a within-day and a within-week cycle, and extended Holt-Winters exponential smoothing to capture both, outperforming a double seasonal ARIMA model. Such multiple seasonalities can also be represented with Fourier terms in a regression framework, an approach that handles long and non-integer seasonal periods flexibly [5].

Machine learning methods based on decision trees have become widely used because they capture nonlinear interactions between predictors without requiring them to be specified in advance. Random forests average many regression trees grown on bootstrap samples to reduce variance [6], while gradient boosting builds trees sequentially, each correcting the errors of the previous ones; XGBoost is a regularised and highly efficient implementation of this idea [7]. In a tutorial review, Hong and Fan [2] emphasised that new forecasting methods should be evaluated against appropriate benchmarks on reproducible case studies, a principle followed in this study.

2.2 Electricity demand forecasting in South Africa

South African research has focused mainly on daily peak demand. Sigauke and Chikobvu [8] compared SARIMA, SARIMA-GARCH and a regression-SARIMA-GARCH model whose predictors were selected by multivariate adaptive regression splines, and found that the latter forecast daily peak demand most accurately, with a MAPE of 1.42%. Chikobvu and Sigauke [9] modelled daily peak demand from 1996 to 2009 with SARIMA and regression-SARIMA models, and found that the SARIMA model gave the more accurate short-term forecasts, while the regression variant captured the main drivers of demand.

2.3 Research gap

These South African studies forecast daily peak demand, use data from well before the recent load shedding crisis and apply classical time-series models. Few studies evaluate tree-based machine learning methods on hourly national demand in the period after 2022, which is shaped by severe load shedding and growing embedded solar generation, and few compare such models directly with the system operator's published forecast. This study addresses that gap.

3 Data and pre-processing

3.1 Data source

Hourly data were obtained from the Eskom Data Portal [10] through its data request form, which provides a download link by email. All Demand Side series were requested, together with two Supply Side series indicating load reduction, for the period 1 April 2022 to 31 March 2027. Table 1 lists the variables used.

Table 1: Variables used in the study.

Variable	Description	Role in this study
Date Time Hour Beginning	Start of each hourly interval	Time index
RSA Contracted Demand	Hourly demand that Eskom must supply in South Africa (MW)	Target variable
RSA Contracted Forecast	Eskom's published forecast of RSA contracted demand (MW)	Benchmark
Manual Load_Reduction (MLR)	Demand removed through load shedding (MW)	Load shedding indicator
ILS Usage	Interruptible load supply dispatched (MW)	Exploratory analysis

3.2 Pre-processing

All processing was carried out in Python using pandas [11]. Leading spaces were removed from column names, and timestamps recorded on a 12-hour clock were parsed with the explicit format `%Y-%m-%d %I:%M:%S %p` to avoid confusing midnight with midday. The series was then checked for unreadable timestamps, duplicate hours and gaps against a complete hourly calendar, and demand values outside the range 10,000 to 40,000 MW were treated as erroneous.

The source file contained 43,824 rows, but observed demand ended on 2 November 2026. The remaining 3,576 rows held only Eskom's forward forecasts to the end of the financial year and were excluded. No unreadable timestamps, duplicate hours, missing hours or out-of-range values were found, so no imputation was required. The final dataset comprised 40,248 consecutive hourly observations from 1 April 2022 to 2 November 2026.

4 Methodology

4.1 Problem formulation

Let y_t denote RSA contracted demand (MW) in hour t. The task is to produce a forecast of y_t using only information available at least 24 hours earlier:

$$\hat{y}_t = f(\mathbf{x}_t), \qquad \mathbf{x}_t \in \mathcal{F}_{t-24}, \tag{1}$$

where $\mathbf{x}_t$ is a vector of predictors and $\mathcal{F}_{t-24}$ denotes the information available at time $t-24$.

4.2 Predictors

The daily, weekly and annual cycles identified in the exploratory analysis were represented by Fourier terms [5]. For a cycle of period P hours and harmonic k,

$$s_k(t) = \sin\!\left(\frac{2\pi k t}{P}\right), \qquad c_k(t) = \cos\!\left(\frac{2\pi k t}{P}\right), \tag{2}$$

with $P = 24$ and $k = 1, \ldots, 4$ for the daily cycle, $P = 168$ and $k = 1, 2, 3$ for the weekly cycle, and $P = 8{,}766$ and $k = 1, 2$ for the annual cycle. To allow the shape of the daily cycle to vary with the season, each daily term was also multiplied by the first annual sine and cosine terms. The full set of predictors is summarised in Table 2.

Table 2: Predictors used in the models.

Group	Predictors
Calendar	Hour of day, day of week, month, weekend indicator, South African public holiday indicator, day after a public holiday, festive season indicator (20 December to 6 January)
Load shedding	Indicator equal to 1 when manual load reduction was active
Fourier terms	Daily (8 terms), weekly (6 terms) and annual (4 terms), from Equation (2)
Season × hour	Products of the 8 daily terms with the first annual sine and cosine (16 terms)
Lagged demand	y_{t-24}, y_{t-168}, and the mean of y over the 168 hours ending at $t-24$

4.3 Models

Naive benchmarks. Two benchmarks were used: the same hour on the previous day and the same hour in the previous week,

$$\hat{y}_t^{\mathrm{N1}} = y_{t-24}, \qquad \hat{y}_t^{\mathrm{N2}} = y_{t-168}. \tag{3}$$

Fourier regression. Demand was modelled as a linear function of the Fourier terms, season × hour products, calendar and load shedding indicators, and the lagged 7-day mean,

$$y_t = \beta_0 + \sum_{j=1}^{p} \beta_j \, x_{j,t} + \varepsilon_t, \tag{4}$$

with the coefficients estimated by ordinary least squares,

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 = \left(\mathbf{X}^\top \mathbf{X}\right)^{-1} \mathbf{X}^\top \mathbf{y}. \tag{5}$$

The estimates obtained with scikit-learn were verified by solving Equation (5) directly with NumPy's least squares routine; both gave the same coefficients.

Random forest. The random forest prediction is the average of B regression trees T_b, each grown on a bootstrap sample of the training data [6]:

$$\hat{y}_t = \frac{1}{B} \sum_{b=1}^{B} T_b(\mathbf{x}_t). \tag{6}$$

$B = 200$ trees were used, with a minimum of three observations per leaf; following the scikit-learn default for regression, all predictors were considered at each split.

XGBoost. Gradient boosting builds an additive model of K trees f_k, fitted sequentially to minimise a regularised objective [7]:

$$\mathcal{L} = \sum_t (y_t - \hat{y}_t)^2 + \sum_{k=1}^{K} \Omega(f_k), \qquad \Omega(f) = \gamma T + \tfrac{1}{2}\lambda \|\mathbf{w}\|^2, \tag{7}$$

where T is the number of leaves and $\mathbf{w}$ the leaf weights of a tree. The model used $K = 800$ trees, a learning rate of 0.03, a maximum depth of 6, and 80% row and column subsampling. The random forest and XGBoost used the calendar, load shedding, Fourier and lagged-demand predictors; the season $\times$ hour products were used only in the linear model, since tree models learn such interactions directly.

4.4 Experimental design

The data were split chronologically. All models were evaluated on a hold-out test period from 1 May to 2 November 2026 (4,464 hours), which includes the 2026 winter and was never used for estimation. To assess the effect of load shedding on estimation, each model was trained under two settings:

- **Option A (full sample):** April 2022 to 30 April 2026, 35,593 hours after removing the first 191 hours, for which the lagged predictors were unavailable.
- **Option B (post-load shedding sample):** 1 April 2024 to 30 April 2026, 18,240 hours.

Eskom's published forecast was evaluated over the same test hours.

4.5 Evaluation metrics

Forecast accuracy was measured with the mean absolute error (MAE), root mean squared error (RMSE) and mean absolute percentage error (MAPE):

$$\mathrm{MAE} = \frac{1}{n}\sum_{t=1}^{n} |y_t - \hat{y}_t|, \quad \mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{t=1}^{n} (y_t - \hat{y}_t)^2}, \quad \mathrm{MAPE} = \frac{100\%}{n}\sum_{t=1}^{n} \left|\frac{y_t - \hat{y}_t}{y_t}\right|. \tag{8}$$

Because errors at the evening peak have the greatest operational consequences, the MAE was also computed for the hours 17:00 to 20:59 (peak MAE). All models were implemented in Python using scikit-learn [12] and the xgboost library [7].

5 Results

5.1 Descriptive statistics

Hourly demand over the full period averaged 24,650 MW, with a range of 16,621 to 34,666 MW (Table 3).

Table 3: Descriptive statistics of hourly RSA contracted demand, 1 April 2022 to 2 November 2026.

Statistic	Value (MW)
Observations (hours)	40,248
Mean	24,650
Standard deviation	3,109
Minimum	16,621
First quartile	22,067
Median	24,741
Third quartile	27,077
Maximum	34,666

5.2 Exploratory analysis

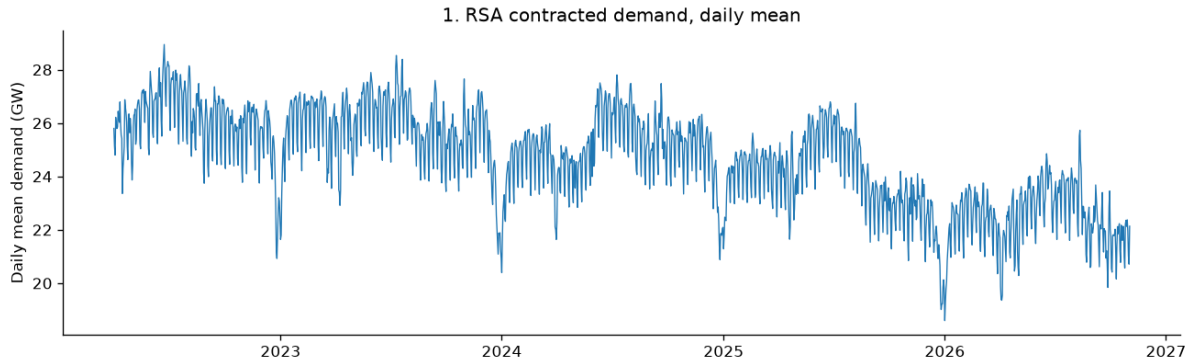


Figure 1: Daily mean RSA contracted demand, 1 April 2022 to 2 November 2026.

Figure 1 shows a weekly cycle, with lower demand at weekends; an annual cycle with peaks of about 27 to 28 GW each June to August in 2022 to 2024; and sharp falls over each December festive season, to a low of about 18.6 GW in January 2026. Typical daily demand declined from about 26 to 27 GW in 2022 to about 22 GW in 2026, with an apparent step decrease around September 2025. This decline is consistent with the growth in embedded solar generation reported by the CSIR [3].

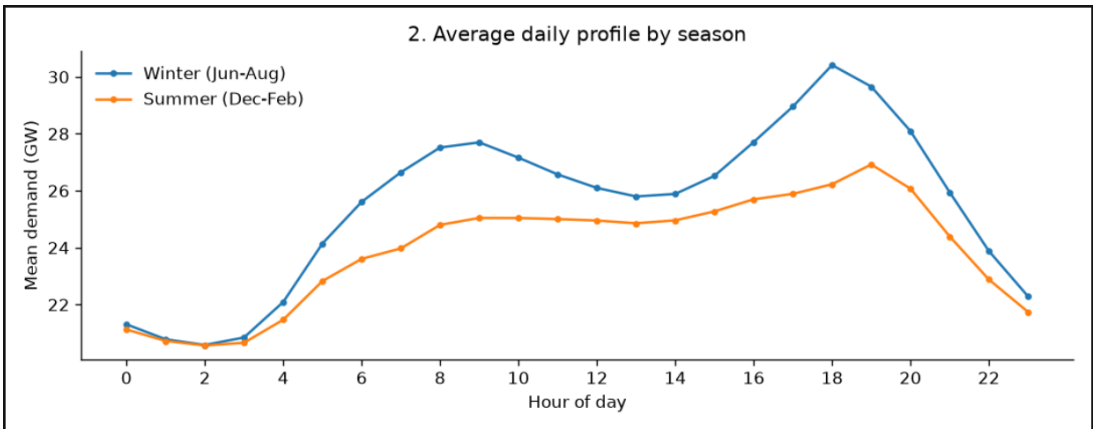


Figure 2: Mean demand by hour of day, winter (June to August) and summer (December to February).

Night-time demand was similar in both seasons, at about 20.6 GW around 02:00 (Figure 2). In winter, demand rose to a morning peak of about 27.7 GW at 09:00, fell to about 25.8 GW at 13:00 and reached an evening peak of 30.4 GW at 18:00. The summer evening peak was lower, at 26.9 GW, and occurred an hour later, at 19:00.

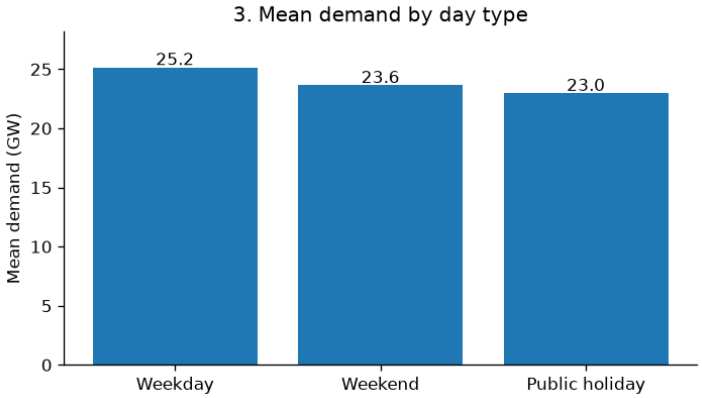


Figure 3: Mean demand on weekdays, weekends and public holidays.

Mean demand was 25.2 GW on weekdays, 23.6 GW at weekends and 23.0 GW on public holidays, which are 6% and 9% below the weekday level respectively (Figure 3).

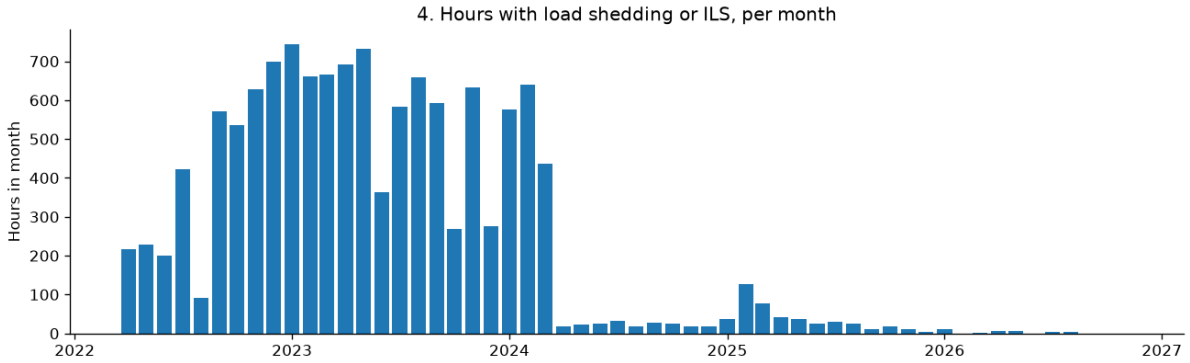


Figure 4: Hours per month in which manual load reduction or interruptible load supply (ILS) was active.

Manual load reduction or ILS was active in 12,799 hours (31.8% of the sample). These hours

were concentrated between April 2022 and March 2024, at 200 to 750 hours per month, after which they fell to near zero apart from a small increase in early 2025 (Figure 4). Because ILS is also dispatched outside load shedding, this proportion overstates the incidence of load shedding itself.

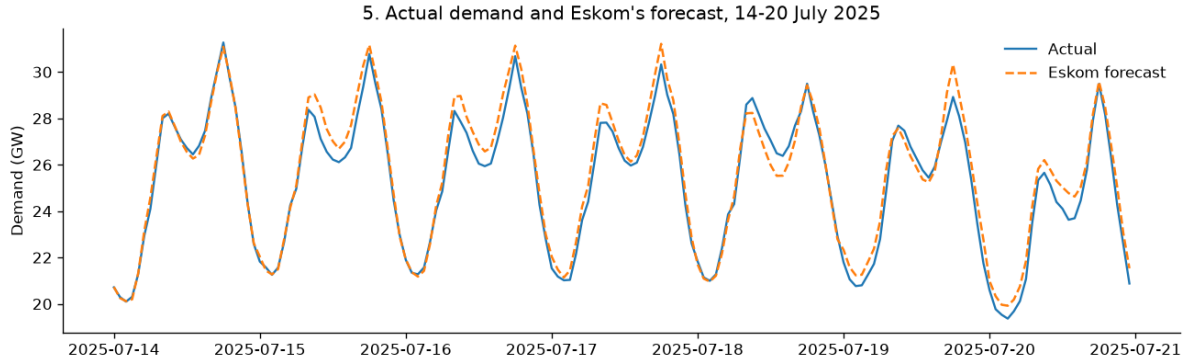


Figure 5: Actual demand and Eskom's forecast, 14 to 20 July 2025.

Over the full sample, Eskom's published forecast had an MAE of 419 MW and a MAPE of 1.72%. It reproduced both daily peaks closely but overestimated several evening peaks and the Sunday night-time trough (Figure 5).

5.3 Forecast accuracy

Table 4 reports the accuracy of all models over the test period.

Table 4: Forecast accuracy over the test period, 1 May to 2 November 2026 ($n = 4{,}464$ hours), ordered by MAE. A = full sample; B = sample from April 2024. Best values in bold.

Model	Training sample	MAE (MW)	RMSE (MW)	MAPE (%)	Peak MAE (MW)
XGBoost	B	**522.3**	**723.3**	**2.30**	461.8
Random forest	A	525.3	764.5	2.32	**447.7**
XGBoost	A	538.2	739.3	2.38	466.8
Random forest	B	539.2	787.1	2.38	448.0
Eskom published forecast	–	545.1	730.2	2.40	535.1
Naive, same hour previous day	–	790.2	1,120.5	3.50	687.5
Naive, same hour previous week	–	837.0	1,230.5	3.65	840.9
Fourier regression	B	976.5	1,224.9	4.35	679.1
Fourier regression	A	1,141.3	1,417.8	5.11	675.3

XGBoost trained on sample B achieved the lowest MAE (522.3 MW), RMSE (723.3 MW) and MAPE (2.30%). Relative to the same-hour-previous-day benchmark it reduced the MAE by 33.9%, and relative to Eskom's forecast by 4.2%. The tree-based models were markedly more accurate than Eskom's forecast during the evening peak: peak MAE ranged from 447.7 to 466.8 MW, compared with 535.1 MW for Eskom. The Fourier regression was the least accurate model overall, although its peak MAE was lower than that of both naive benchmarks. Eskom's forecast was less accurate in the test period (MAPE 2.40%) than over the full sample (1.72%).

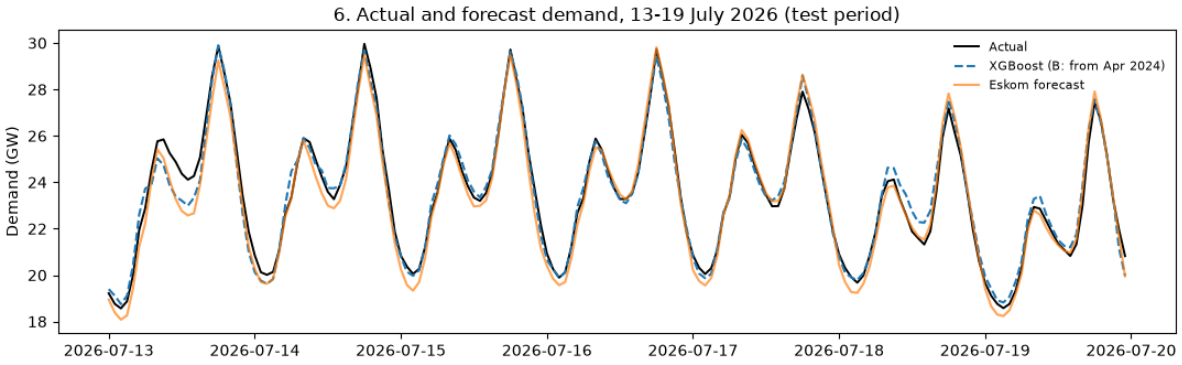


Figure 6: Actual demand, the XGBoost (sample B) forecast and Eskom's forecast, 13 to 19 July 2026.

Figure 6 illustrates a winter week from the test period. The XGBoost forecast tracks the evening peaks of close to 30 GW, whereas Eskom's forecast underestimates several night-time troughs. XGBoost overestimates the weekend morning peaks on 18 and 19 July, and neither forecast captures the elevated late-morning demand on 13 July.

5.4 Predictor importance

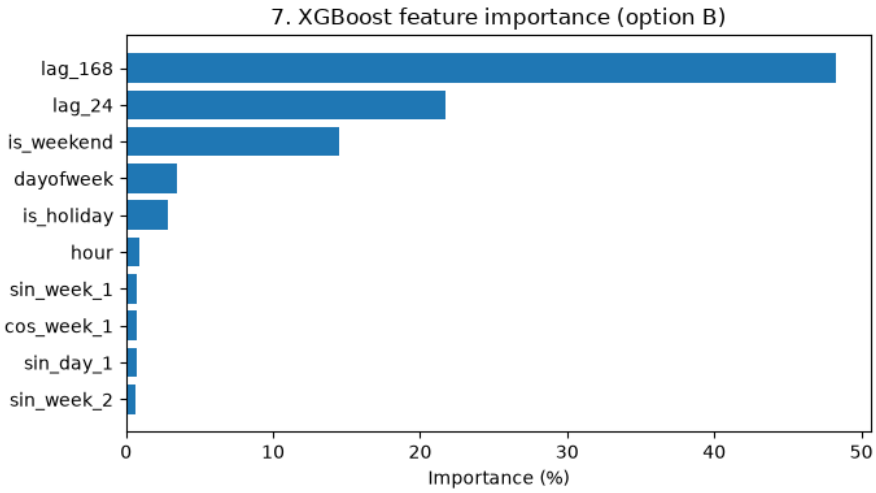


Figure 7: The ten most important predictors in the XGBoost model (sample B).

Demand in the same hour one week earlier accounted for about 48% of the total importance, followed by demand 24 hours earlier (about 22%) and the weekend indicator (about 14%). Day of week and the public holiday indicator each contributed about 3%, and every remaining predictor less than 1% (Figure 7).

6 Discussion

6.1 Forecast accuracy

The best model forecast hourly demand 24 hours ahead with a MAPE of 2.30% and an MAE of 522 MW, about 2% of mean demand. This level of accuracy is broadly in line with published South African work: Sigauke and Chikobvu [8] reported a MAPE of 1.42% for daily peak demand. The two results are not directly comparable, however, since forecasting 24 separate hourly values is a harder task than forecasting a single daily maximum.

6.2 Comparison with benchmarks and Eskom's forecast

All tree-based models clearly outperformed the naive benchmarks, reducing the MAE by about a third. Consistent with Hong and Fan [2], this confirms that the models learn structure beyond simple repetition of past demand.

Against Eskom's published forecast, the best model achieved a 4.2% lower MAE and a slightly lower MAPE (2.30% against 2.40%). Two considerations limit how strongly this can be interpreted. First, the models were evaluated as rolling 24-hour-ahead forecasts, in which the forecast for each hour uses demand from exactly 24 hours earlier. An operational day-ahead forecast is issued once for all hours of the following day, and therefore cannot use the most recent observations; the horizon at which Eskom's published forecast was produced is not stated in the data. Second, the differences between the four best models and Eskom's forecast lie within 0.10 percentage points of MAPE over a single six-month test period. The results therefore show that the models performed *comparably* to Eskom's forecast overall.

The clearest advantage was at the evening peak, where the tree-based models reduced the MAE by 68 to 87 MW, up to 16%, relative to Eskom's forecast. Eskom's forecast also deteriorated from a MAPE of 1.72% over the full sample to 2.40% in the test period, and overestimated demand in several of the periods examined. Together with the decline in demand shown in Figure 1, this suggests that the operator's forecast has not fully adjusted to the reduction in grid demand associated with embedded solar generation [3], whereas models that rely on recent observed demand adapt to it automatically.

6.3 The effect of load shedding data

Restricting training to the period after April 2024 improved XGBoost (MAPE 2.30% against 2.38%) and the Fourier regression (4.35% against 5.11%), but not the random forest (2.38% against 2.32%). The benefit was largest for the linear model, whose single set of coefficients cannot represent both the period of suppressed demand and the period after it. The flexible tree models were less affected. The evidence therefore suggests that excluding the load shedding period is helpful for simpler models, but not consistently necessary for tree-based models.

6.4 Important predictors

Lagged demand dominated the XGBoost model: demand one week earlier and 24 hours earlier together accounted for about 70% of the importance, followed by the weekend and holiday indicators. The model therefore operates largely by adapting last week's demand pattern to calendar effects and recent levels, which explains its robustness to the downward trend. The

Fourier terms contributed little to the tree models, as the same information is available from the hour and day-of-week predictors.

The poor performance of the Fourier regression has a clear explanation: it was the only model without demand from 24 hours or one week earlier, relying instead on a 7-day average for the current level. It captured the average daily shape, as shown by its peak MAE below that of the naive benchmarks, but could not respond to short-term changes in demand.

6.5 Limitations

- Weather variables, particularly temperature, which drives the winter evening peak, were not included.
- The load shedding indicator is based on recorded load reduction, whereas announced load shedding stages may change at short notice. Since load shedding was rare during the test period, this has little effect on the reported results.
- Hyperparameters were set by judgement rather than tuned by cross-validation.
- The evaluation covers a single test period, and the cause of the apparent step decrease in demand around September 2025 was not established.

7 Conclusion and recommendations

This study evaluated statistical and machine learning methods for forecasting South Africa's hourly electricity demand 24 hours ahead, using 40,248 hours of Eskom data from April 2022 to November 2026. The main conclusions are as follows.

1. Tree-based machine learning models forecast hourly demand with a MAPE of about 2.3%, reducing the error of a naive same-hour-previous-day forecast by about a third.
2. The best model, XGBoost trained on data from April 2024, performed comparably to Eskom's published forecast overall (MAPE 2.30% against 2.40%), and was up to 16% more accurate during the evening peak.
3. Excluding the period of heavy load shedding improved the linear model substantially and XGBoost slightly, but not the random forest.
4. Lagged demand, particularly from one week earlier, was the most important predictor, allowing the models to adapt to the marked decline in grid demand between 2022 and 2026.

Recommendations for further research.

1. Repeat the evaluation as a strict day-ahead forecast, using only information available at a fixed issue time on the previous day, to allow a fully like-for-like comparison with Eskom's forecast.
2. Evaluate the models over several test periods, for example with rolling-origin cross-validation, and test the significance of accuracy differences.
3. Add lagged demand to the Fourier regression and include temperature data to improve forecasts of the winter evening peak.
4. Tune model hyperparameters with time-series cross-validation.
5. Investigate the causes of the decline in grid demand, including embedded solar generation, and model it explicitly.
6. Extend the work to probabilistic forecasts that quantify uncertainty, as advocated by Hong and Fan [2].

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